Jrenoth Misquith

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ACHIEVEMENTS

- **100% Academic Merit Scholarship Recipient** (140,000 HKD p.a) & Dean's List Semester A 2023/24
- **Ranked 140th globally** in **Bloomberg Equity Trading Competition** (2024)
- **HKTECH300 & HKSTP Seed Fund** recipient for MuGen (Music ML startup)
- **CityHack finalist (2nd Runner Up)** and Most Creative Team Award winner
- Generated **10x returns** on Indian Market investments (June 2022-2024)
- Passed the **Hyrotrader challenge**, achieving 13% profit within one month through cryptocurrency futures trading and strict risk management (max 3% portfolio risk per trade with **Backtesting results**)
- **Microsoft Student Ambassador** (Oct '23 – Feb '24) with **Azure AI Builder** certifications
- **Executive Council Member –** CityU English Debate Team (2021-'23) & Model UN Team (2022-'24)
- **Volunteer** at **Make A Difference-India**, an NGO dedicated to underprivileged children's education.

WORK EXPERIENCE

Banza **Jun '25 – Oct '25**
Backend Developer *Dubai (Remote)*

- As a freelance backend developer, I was involved in engineering the core AI-driven backend for a movie recommendation engine. Architected and deployed a scalable, event-driven microservices system that aggregated user data from 8+ entertainment and social platforms to deliver hyper-personalized movie suggestions with critic-style explanations. Applied Postman and FastAPI for creating routes for user data and mostly did server-side development with Python. Tried replicating the frontend with React and Express.js.

Impact: System Performance & Scalability: The designed backend successfully processed data from over 10,000 simulated user profiles, with minimal downtime. The implemented PostgreSQL analytics layer equipped the product team with actionable insights, allowing for the initial tracking of key metrics.

Mycoach Health **Jun '23 – Sep '23**
Software Engineer & Data Science Intern *HKSTP, Hong Kong*

- Collaborated with Co-founder to fine-tune LLMs for healthcare applications
- Developed Health GPT model for medical data storage and automated inference generation. Picked up NoSQL concepts like Vector DBs (like Redis), MongoDB and searches alongwith version control and analyzing git workflows.

Impact: Improved data retrieval efficiency by 40% and enabled automated analysis of patient reports, reducing doctor review time by approximately 25%.

PricewaterhouseCoopers **Sep '23 – Mar '24**
Software Engineering Placement Trainee *Central, Hong Kong*

- Led intern initiatives across teams for website builders and Salesforce API integrations with Java.
- Designed Azure-based architectures to optimize system performance and operations. Worked with Power BI to make interactive analytics dashboards and presentations towards the client-side.

Impact: Reduced development time for client websites by 30% and improved HR service delivery speed and important analytics like engaging customer traffic for clients and their project needs.

EDUCATION

College of Engineering, City University of Hong Kong (Top 100 Ranked University) | *2021-2025*
Major: BSc. Computer Science with specialization in Data Science | **CGPA:** 3.32/4.30
Minor: Finance - Focus on financial markets, backend development and A.I

PROJECTS

Hospital Patient Portal (Fullstack AI development)
-**Architected and delivered "Patient Portal"**, a high-performance Progressive Web Application built with React 19 frontend designed to modernize patient interaction with a complex legacy backend system.
- Achieved **sub-2 second load times** and a **60% improvement in UI responsiveness** by migrating from legacy server-side rendering to a modern Single Page Application (SPA) architecture.
- Reduced administrative overhead by **~30%** by digitizing previously manual workflows, including self-service appointment booking, vital tracking, and instant lab report retrieval. Made the UI intuitive, while including 3D designs, and accommodated/made extendible for AI chatbot integration.
- Secured **100% of digital revenue streams** by engineering a fault-tolerant Razorpay payment gateway for Pharmacy and Appointment modules, handling thousands of transactions without data loss.
- Increased patient data accessibility by **100%**, enabling secure, real-time "Health Bubble" messaging between patients and care teams that previously required physical visits.

AI Automated Movie Recommender (Backend, Agentic AI and Data Extraction)
-Built an AI agentic workflow pipeline connecting **MongoDB** data stores to raw user platform data (Twitter/Amazon Prime/Spotify) to extract and store user preferences using structured **Pydantic** models.
-Developed automated data processing pipelines for 8+ social media platforms (Twitter, Instagram, YouTube, Twitch, Steam, LinkedIn, Netflix, Hotstar) with **MongoDB** storage and preference document generation.
-Created an intelligent recommendation engine using TMDB API integration with AI agents that generate percentage-based movie ratings and movie critic-style explanations for personalized recommendations.
-Implemented automated Git workflow with **Model Context Protocol (MCP)** and **PostgreSQL** storage for structured recommendation data with user engagement tracking and analytics.

Music Generation App (MuGEN, HKSTP Ideathon): Startup Idea
-Developed an application that allows musicians to use gestures with their fingers to control the editing of their music/sound tracks.
-Refined and innovated on the music understanding for the models, plugged in from Google's Magenta project. Applied **Figma, Vanilla JavaScript** and **AI plugins** to build the frontend. Offset music storage compute processes to the cloud using Azure and trained upto **Microsoft AZ900** during user trials.
-Built a dashboard that takes the user from A to Z of importing their tracks, enhancing/prolonging their sound further with the use of AI. Additionally, recognized best project in my course: Multimodal Design of Interfaces.

SKILLS

- **Languages:** English (Native, IELTS: 8.5/9.0), Kannada (Native), Konkani (Native), Hindi (Conversational)
- **Technical:** Python, Bloomberg Terminal, Pandas, NumPy, Tableau, PyTorch, Java, Spring, Azure, Excel, Git, FastAPI, Databases (SQL: PostgresDB, MySQL & NoSQL: MongoDB) & Figma
- **Finance:** Financial Data Analysis, Technical Analysis, Business Cycle Analysis, Elliott Wave Theory & Macroeconomics